

CSS Writing-Mode

Complete Guide: Text Direction and Orientation

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1 Introduction

The CSS `writing-mode` property controls the text direction and orientation on web pages. It's essential for supporting multiple languages, scripts, and design effects that require non-horizontal text flow.

1.1 Text Direction Basics

Different languages require different text directions:

- **Horizontal:** English, French, German, most European languages
- **Right-to-Left:** Arabic, Hebrew, Persian, Urdu
- **Vertical:** Chinese, Japanese, Korean (traditional writing)
- **Vertical Right-to-Left:** Some East Asian scripts

1.2 Why `writing-mode` Matters

- Support for multilingual content
- Proper text layout for all scripts
- Creative vertical text effects
- Appropriate handling of CJK languages
- Accessible design for all users
- Cultural and linguistic respect

2 writing-mode Property Values

2.1 Available Values

Value	Direction	Use	Languages
<code>horizontal-tb</code>	Left to right, top to bottom	Default	LTR
<code>vertical-rl</code>	Top to bottom, right to left	East Asian	CJK
<code>vertical-lr</code>	Top to bottom, left to right	East Asian	CJK
<code>sideways-rl</code>	Rotated right	Decorative	All
<code>sideways-lr</code>	Rotated left	Decorative	All

3 writing-mode: horizontal-tb

The default writing mode. Text flows left-to-right, top-to-bottom.

3.1 Behavior

- Standard horizontal text layout
- Text reads left to right
- New lines appear below
- Default for most European languages
- Most common use case

3.2 Examples

```
/* Default (explicitly set) */
body {
  writing-mode: horizontal-tb;
}

/* English text */
p {
  writing-mode: horizontal-tb;
}

/* Equivalent to normal text flow */
div {
  writing-mode: horizontal-tb;
}
```

4 writing-mode: vertical-rl

Text flows top-to-bottom, right-to-left. Common for traditional East Asian writing.

4.1 Behavior

- Text flows vertically downward
- Lines progress from right to left
- Common in traditional Chinese/Japanese
- Text rotates 90 degrees clockwise
- Paragraph columns read right-to-left

4.2 Examples

```
/* Traditional Chinese */
.traditional-chinese {
  writing-mode: vertical-rl;
  font-family: SimSun, serif;
}

/* Traditional Japanese */
.traditional-japanese {
  writing-mode: vertical-rl;
  font-family: 'Hiragino Mincho Pro', serif;
}

/* Article in vertical layout */
article {
  writing-mode: vertical-rl;
  max-height: 500px;
  overflow-y: auto;
}
```

5 writing-mode: vertical-lr

Text flows top-to-bottom, left-to-right. Alternative vertical orientation.

5.1 Behavior

- Text flows vertically downward
- Lines progress from left to right
- Less common than vertical-rl
- Rotates text 90 degrees counter-clockwise
- Used in some vertical scripts

5.2 Examples

```
/* Alternative vertical layout */
.vertical-left {
  writing-mode: vertical-lr;
}

/* Mongolian script */
.mongolian {
  writing-mode: vertical-lr;
  font-family: Mongolian Baiti, serif;
}

/* Creative vertical text */
.vertical-design {
  writing-mode: vertical-lr;
  text-orientation: mixed;
}
```

6 writing-mode: sideways-rl

Text is rotated 90 degrees clockwise, reading sideways from right to left.

6.1 Behavior

- Text rotates 90 degrees clockwise
- Characters remain upright in rotation
- All characters point right
- Limited browser support
- Experimental feature

6.2 Examples

```
/* Sideways text (right-to-left) */
.sideways-right {
  writing-mode: sideways-rl;
}

/* Experimental rotated text */
.rotated {
  writing-mode: sideways-rl;
  font-size: 18px;
}

/* Note: Limited browser support */
.experimental {
  writing-mode: sideways-rl;
  /* May not work in all browsers */
}
```

```
}
```

7 writing-mode: sideways-lr

Text is rotated 90 degrees counter-clockwise, reading sideways from left to right.

7.1 Behavior

- Text rotates 90 degrees counter-clockwise
- Characters remain upright in rotation
- All characters point left
- Very limited browser support
- Experimental feature

7.2 Examples

```
/* Sideways text (left-to-right) */  
.sideways-left {  
  writing-mode: sideways-lr;  
}  
  
/* Alternative rotation */  
.rotated-left {  
  writing-mode: sideways-lr;  
}  
  
/* Experimental - limited support */
```

8 Real-World Use Cases

8.1 Use Case 1: Multilingual Content

```
/* Conditional language support */
html[lang="zh"] {
  writing-mode: vertical-rl;
}

html[lang="ja"] {
  writing-mode: vertical-rl;
}

html[lang="en"] {
  writing-mode: horizontal-tb;
}

html[lang="ar"] {
  writing-mode: horizontal-tb;
  direction: rtl;
}

/* Language-specific styling */
.content-zh {
  writing-mode: vertical-rl;
  font-family: SimSun, serif;
}

.content-en {
  writing-mode: horizontal-tb;
  font-family: Georgia, serif;
}

.content-ar {
  writing-mode: horizontal-tb;
  direction: rtl;
  font-family: Arial, sans-serif;
}
```

8.2 Use Case 2: Creative Vertical Headings

```
/* Vertical title */
.title-vertical {
  writing-mode: vertical-rl;
  font-size: 32px;
  font-weight: 700;
  margin: 0 0 20px 0;
}

/* Sidebar heading */
.sidebar-title {
  writing-mode: vertical-rl;
}
```



```
    text-orientation: mixed;
    padding: 10px;
    background-color: #f0f0f0;
}

/* Portfolio header */
.portfolio-title {
    writing-mode: vertical-rl;
    font-size: 24px;
    text-decoration: underline;
}
```

8.3 Use Case 3: Table Headers

```
/* Vertical table headers */
th {
    writing-mode: vertical-rl;
    padding: 10px;
    height: 200px;
}

/* Rotate text for compact tables */
table.compact th {
    writing-mode: vertical-rl;
    text-orientation: mixed;
}

/* Report headers */
.report-header {
    writing-mode: vertical-rl;
    font-weight: 600;
}
```

8.4 Use Case 4: Vertical Layout Design

```
/* Full vertical layout */
body {
    writing-mode: vertical-rl;
    display: flex;
    flex-direction: row-reverse;
}

article {
    writing-mode: vertical-rl;
    max-height: 600px;
    overflow-y: auto;
    padding: 20px;
}

/* Column layout */
.column {
```

```

    writing-mode: vertical-rl;
    width: 200px;
    margin-right: 20px;
}

.column:first-child {
    border-right: 1px solid #ddd;
}

```

8.5 Use Case 5: East Asian Typography

```

/* Chinese article */
article.chinese {
    writing-mode: vertical-rl;
    font-family: SimSun, serif;
    line-height: 1.8;
    font-size: 16px;
}

/* Japanese manga-style */
.manga {
    writing-mode: vertical-rl;
    text-orientation: upright;
    font-family: 'YuGothic', sans-serif;
}

/* Korean with alternatives */
.korean {
    writing-mode: horizontal-tb; /* Modern Korean uses horizontal */
    direction: ltr;
}

/* Traditional Korean (rare) */
.korean-traditional {
    writing-mode: vertical-rl;
}

```

9 text-orientation Property

The `text-orientation` property controls how characters are rotated in vertical writing modes.

9.1 text-orientation Values

Value	Behavior	Use
<code>mixed</code>	Rotate some characters	Default vertical
<code>upright</code>	Keep all characters upright	Preserve orientation
<code>sideways</code>	Rotate all characters 90°	Experimental

9.2 Examples

```

/* Default mixed orientation */
.vertical {
  writing-mode: vertical-rl;
  text-orientation: mixed;
}

/* All characters upright */
.uptight {
  writing-mode: vertical-rl;
  text-orientation: upright;
}

/* Sideways rotation */
.sideways {
  writing-mode: vertical-rl;
  text-orientation: sideways;
}

/* Numbers and Latin stay horizontal */
.mixed-text {
  writing-mode: vertical-rl;
  text-orientation: mixed;
}

/* Japanese with mixed orientation */
.japanese-mixed {
  writing-mode: vertical-rl;
  text-orientation: mixed;
}

/* Japanese with upright characters */
.japanese-upright {
  writing-mode: vertical-rl;
  text-orientation: upright;
}

```

10 Browser Support

10.1 writing-mode Support

Value	Chrome	Firefox	Safari	Edge
horizontal-tb	All	All	All	All
vertical-rl	All	All	All	All
vertical-lr	All	41+	10.1+	All
sideways-rl	91+	No	No	91+
sideways-lr	91+	No	No	91+

10.2 Fallback Strategies

```

/* Provide fallback for older browsers */

```

```
.vertical-text {
  writing-mode: vertical-rl;
  writing-mode: tb-rl; /* Legacy IE */

  /* Fallback for no support */
  transform: rotate(-90deg); /* Not ideal, but works */
}

/* Feature detection */
@supports (writing-mode: vertical-rl) {
  .vertical {
    writing-mode: vertical-rl;
  }
}

@supports not (writing-mode: vertical-rl) {
  .vertical {
    /* Use alternative method */
    transform: rotate(-90deg);
    transform-origin: top left;
  }
}
```

11 Combining writing-mode with Other Properties

11.1 writing-mode and width/height

```
/* Vertical text with height constraint */
.vertical-box {
  writing-mode: vertical-rl;
  height: 300px;
  width: 50px;
  overflow-y: auto;
}

/* Vertical text with padding */
.padded-vertical {
  writing-mode: vertical-rl;
  padding: 20px 10px;
}

/* Fixed dimensions */
.vertical-container {
  writing-mode: vertical-rl;
  width: 100px;
  height: 400px;
  border: 1px solid #ddd;
}
```

11.2 writing-mode and Flexbox

```
/* Vertical flex layout */
.flex-vertical {
  display: flex;
  flex-direction: column;
}

.flex-vertical .item {
  writing-mode: vertical-rl;
  flex: 1;
}

/* Row of vertical columns */
.flex-row {
  display: flex;
  flex-direction: row;
}

.flex-row .column {
  writing-mode: vertical-rl;
  flex: 1;
  border-right: 1px solid #ddd;
}
```

11.3 writing-mode and Grid

```
/* Grid with vertical text */
.grid-vertical {
  display: grid;
  grid-template-columns: auto auto auto;
  gap: 10px;
}

.grid-vertical .cell {
  writing-mode: vertical-rl;
  padding: 10px;
}

/* 2x2 grid */
.grid-2x2 {
  display: grid;
  grid-template-columns: 1fr 1fr;
  grid-template-rows: auto auto;
}

.grid-2x2 .item {
  writing-mode: vertical-rl;
  border: 1px solid #ddd;
}
```

11.4 writing-mode and Transform

```
/* Rotate vertical text further */
.rotated-vertical {
  writing-mode: vertical-rl;
  transform: rotate(180deg);
}

/* Flip vertical text */
.flipped {
  writing-mode: vertical-rl;
  transform: scaleX(-1);
}

/* Mirror effect */
.mirrored {
  writing-mode: vertical-rl;
  transform: scaleY(-1);
}
```

12 Multilingual Considerations

12.1 Language-Specific Handling

```
/* HTML attribute selector */
html[lang="zh-TW"] {
    writing-mode: vertical-rl;
}

html[lang="zh-CN"] {
    writing-mode: horizontal-tb; /* Simplified Chinese typically horizontal */
}

html[lang="ja"] {
    writing-mode: vertical-rl;
}

html[lang="ko"] {
    writing-mode: horizontal-tb; /* Modern Korean is horizontal */
}

html[lang="ar"] {
    writing-mode: horizontal-tb;
    direction: rtl;
}

html[lang="he"] {
    writing-mode: horizontal-tb;
    direction: rtl;
}
```

12.2 Mixed Language Content

```
/* Container with specific language */
.content-area {
    writing-mode: vertical-rl;
}

/* Embedded English in vertical text */
.english-inline {
    display: inline;
    /* English characters auto-orient */
}

/* Special handling for Latin text */
.latin-text {
    text-orientation: upright;
    font-family: Georgia, serif;
}

/* Mixed orientation */
.mixed-language {
    writing-mode: vertical-rl;
    text-orientation: mixed;
}
```

13 Common Issues and Solutions

13.1 Issue 1: Text Direction Not Working

Problem:

```
.vertical {  
  writing-mode: vertical-rl;  
  /* Text still appears horizontal */  
}
```

Solution:

```
.vertical {  
  writing-mode: vertical-rl;  
  height: 300px; /* Need height constraint */  
  width: 50px;  
}
```

13.2 Issue 2: Text Overflow

Problem:

```
.vertical-box {  
  writing-mode: vertical-rl;  
  width: 50px;  
  /* Text overflows horizontally */  
}
```

Solution:

```
.vertical-box {  
  writing-mode: vertical-rl;  
  width: 50px;  
  overflow-y: auto; /* Scroll vertical (physical) */  
}
```

13.3 Issue 3: Character Orientation Wrong

Problem:

```
.vertical-text {  
  writing-mode: vertical-rl;  
  /* Numbers appear rotated incorrectly */  
}
```

Solution:

```
.vertical-text {  
  writing-mode: vertical-rl;  
  text-orientation: mixed; /* Correct orientation */  
}
```


13.4 Issue 4: Browser Compatibility

Problem:

```
.vertical {  
  writing-mode: vertical-rl;  
  /* Not supported in older browsers */  
}
```

Solution:

```
.vertical {  
  writing-mode: vertical-rl;  
  
  /* Fallback for unsupported browsers */  
  @supports not (writing-mode: vertical-rl) {  
    transform: rotate(-90deg);  
    transform-origin: top left;  
    display: inline-block;  
  }  
}
```

14 Accessibility Considerations

14.1 Best Practices

- Set `lang` attribute appropriately
- Ensure proper reading order
- Use semantic HTML
- Test with screen readers
- Don't use `writing-mode` for pure decoration
- Provide meaningful content in all directions

14.2 Accessible Examples

```
<!-- Proper lang attribute -->  
<html lang="zh-TW">  
  <body>  
    <article>  
      <!-- Content in Traditional Chinese -->  
      <!-- writing-mode: vertical-rl works correctly -->  
    </article>  
  </body>  
</html>  
  
<!-- Language-specific styling -->  
<article lang="ja" class="vertical">  
  Japanese content in vertical mode  
</article>
```

```
<!-- Accessible vertical heading -->
<h1 class="title-vertical" lang="zh">
  Article Title
</h1>

<!-- Mark-up for clarity -->
<p lang="zh" class="vertical">
  Traditional Chinese text
</p>
```

15 Performance Considerations

15.1 Performance Impact

- `writing-mode` has minimal performance cost
- No layout performance degradation
- Browser handles rendering efficiently
- Multiple vertical elements: no issues
- Works well with modern layout methods

16 Advanced Techniques

16.1 Vertical Navigation

```
/* Vertical navigation menu */
nav {
  writing-mode: vertical-rl;
  display: flex;
  flex-direction: row-reverse;
  gap: 10px;
}

nav a {
  writing-mode: vertical-rl;
  padding: 10px;
  border-bottom: 2px solid transparent;
  transition: border-color 0.3s;
}

nav a:hover {
  border-bottom-color: #3498db;
}
```

16.2 Responsive writing-mode

```
/* Desktop: vertical */
@media (min-width: 1024px) {
  .content {
    writing-mode: vertical-rl;
    height: 600px;
  }
}

/* Tablet: horizontal */
@media (max-width: 1023px) and (min-width: 768px) {
  .content {
    writing-mode: horizontal-tb;
  }
}

/* Mobile: horizontal */
@media (max-width: 767px) {
  .content {
    writing-mode: horizontal-tb;
  }
}
```

17 Summary: Key Takeaways

17.1 writing-mode Values

Value	Direction	Use
horizontal-tb	LTR horizontal	Default
vertical-rl	Top-to-bottom RTL	Traditional CJK
vertical-lr	Top-to-bottom LTR	Alternative vertical
sideways-rl	Rotated right	Experimental
sideways-lr	Rotated left	Experimental

17.2 Best Practices

1. Set `lang` attribute on HTML
2. Use `writing-mode` for language needs
3. Combine with `text-orientation`
4. Test across browsers
5. Provide appropriate dimensions
6. Consider accessibility
7. Don't use purely for decoration
8. Test with screen readers

9. Provide fallbacks for older browsers
10. Use with semantic HTML

17.3 Quick Reference

```
/* Traditional Chinese */
<html lang="zh-TW">
  <body style="writing-mode: vertical-rl;">
    Content
  </body>
</html>

/* Traditional Japanese */
<html lang="ja">
  <article style="writing-mode: vertical-rl;">
    Content
  </article>
</html>

/* Korean (modern - horizontal) */
<html lang="ko">
  <body style="writing-mode: horizontal-tb;">
    Content
  </body>
</html>

/* Arabic (RTL horizontal) */
<html lang="ar">
  <body style="writing-mode: horizontal-tb; direction: rtl;">
    Content
  </body>
</html>

/* Creative vertical heading */
<h1 style="writing-mode: vertical-rl;
          text-orientation: mixed;">
  Vertical Title
</h1>
```

18 Conclusion

The CSS `writing-mode` property is essential for creating truly multilingual web experiences. Supporting different text directions and orientations ensures your web applications serve users worldwide, respecting their languages and cultural preferences.

Key principles to remember:

1. Always set the `lang` attribute
2. Use `vertical-rl` for traditional East Asian
3. Combine with `text-orientation` for proper character placement

4. Test extensively with different languages
5. Consider accessibility and screen readers
6. Use with semantic HTML
7. Provide appropriate dimensions for content
8. Test across browsers
9. Don't rely on older browser support
10. Respect cultural and linguistic diversity

With mastery of writing-mode, you'll create inclusive, globally-accessible web applications that properly support all languages and scripts.

Happy coding!